

Introduction to probability

Probability spaces · Worked student edition

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Errors in a test sample

Suppose a fixed classifier makes 8 errors on 100 tested items. What has been measured?

The observed error fraction is

$$\frac{8}{100} = 0.08.$$

This describes these 100 items. It does not determine the error fraction in the next sample.

Variation in the sampled items can change the measured performance.

A probability question about the same sample

A test sample has 100 items, 8 in error. If one item is selected uniformly, what is the probability it is an error?

Each item has probability $1/100$ of selection.

Eight items are errors, so

$$P(\text{selected item is an error}) = 8 \times \frac{1}{100} = 0.08.$$

Here the probability follows from the stated sampling rule.

A statistical question about future items

Do the 8 errors in the specified sample of 100 establish a future-item error probability of 0.08? State the inference assumption needed.

They provide an estimate, rather than an exact probability for future items.

Using that estimate requires assumptions about how the test sample relates to future items.

Probability starts with a model and calculates consequences. Statistical inference uses observations to learn about a model.

An inspection experiment

Inspect two components in order. Record P for pass and F for fail. List the possible records.

$$\Omega = \{PP, PF, FP, FF\}.$$

PF means that the first component passes and the second fails.

PF and *FP* are different outcomes because the order is recorded.

This list does not yet assign probabilities.

An event as a set of outcomes

For the ordered inspection outcomes PP, PF, FP, FF, list the event A: at least one component fails.

$$A = \{PF, FP, FF\}.$$

The event occurs if the observed outcome belongs to this set.

For example, $PF \in A$, whereas $PP \notin A$.

An event is specified by collecting all outcomes that satisfy the condition.

Exactly one failure

For outcomes PP, PF, FP, FF, let A be at least one failure. Find the event B of exactly one failure and compare it with A .

$$B = \{PF, FP\}, \quad A = \{PF, FP, FF\}.$$

Every outcome in B is also in A , so $B \subseteq A$.

FF belongs to A but not to B .

“At least one” includes two; “exactly one” does not.

Intersection: both conditions

For ordered outcomes PP, PF, FP, FF, let C be first-component failure and D be second-component failure. Find $C \cap D$.

$$C = \{FP, FF\}, \quad D = \{PF, FF\}.$$

The intersection contains the outcomes appearing in both sets:

$$C \cap D = \{FF\}.$$

It describes two failures.

Union: at least one condition

For ordered outcomes PP, PF, FP, FF, let C be first-component failure and D be second-component failure. Find $C \cup D$.

Collect the outcomes appearing in either set, without repetition:

$$C \cup D = \{FP, PF, FF\}.$$

The union describes at least one failure.

The outcome FF is included once.

Complement: the remaining outcomes

For $\Omega = \{PP, PF, FP, FF\}$, let $A = \{PF, FP, FF\}$ be at least one failure. Find A^c .

$$\Omega = \{PP, PF, FP, FF\}, \quad A = \{PF, FP, FF\}.$$

The complement consists of outcomes in Ω that are not in A :

$$A^c = \{PP\}.$$

“No failures” is the complement of “at least one failure.”

Disjoint events

For the ordered inspection outcomes, let $B = \{PF, FP\}$ be exactly one failure and $E = \{FF\}$ be two failures. Determine whether they are disjoint.

$$B = \{PF, FP\}, \quad E = \{FF\}.$$

They have no common outcome:

$$B \cap E = \emptyset.$$

Such events are called disjoint, or mutually exclusive.

Practice: two alarm indicators

Record 1 when an alarm is active, 0 otherwise, and use order for two indicators. List the outcomes in “at least one active,” “exactly one active,” and “neither.”

$$\Omega = \{00, 01, 10, 11\}.$$

At least one active: $\{01, 10, 11\}$.

Exactly one active: $\{01, 10\}$.

Neither active: $\{00\}$.

These descriptions use only sets; they do not require an independence assumption.

Assigning probabilities to outcomes

Suppose the ordered inspection outcomes PP, PF, FP, FF have probabilities 0.72, 0.08, 0.18, 0.02. Check that these are valid outcome probabilities.

Every probability is nonnegative, and

$$0.72 + 0.08 + 0.18 + 0.02 = 1.$$

The values are stipulated model probabilities.

Adding the probabilities in an event

The ordered outcomes PP, PF, FP, FF have probabilities 0.72, 0.08, 0.18, 0.02. Calculate the probability of exactly one failure.

Exactly one failure is $B = \{PF, FP\}$.

$$P(B) = P(\{PF\}) + P(\{FP\}) = 0.08 + 0.18 = 0.26.$$

There are two favourable outcomes out of four, but the answer is not $2/4$.

The four outcomes are not equally likely.

Using the complement

In the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$ and $P(FF)=0.02$. Calculate the probability of at least one failure directly and by the complement.

Direct addition:

$$P(\{PF, FP, FF\}) = 0.08 + 0.18 + 0.02 = 0.28.$$

Complement:

$$1 - P(\{PP\}) = 1 - 0.72 = 0.28.$$

Both calculations refer to the same event.

Deriving the complement rule

Explain why $P(A^c) = 1 - P(A)$.

A and A^c are disjoint and together contain every outcome.

$$P(A) + P(A^c) = P(\Omega) = 1.$$

Rearranging gives

$$P(A^c) = 1 - P(A).$$

Overlapping events in the inspection model

In the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$ and $P(FF)=0.02$. Let C be first-component failure and D second-component failure. Find $P(C)$, $P(D)$ and $P(C \cap D)$.

$$C = \{FP, FF\}, \quad P(C) = 0.18 + 0.02 = 0.20.$$

$$D = \{PF, FF\}, \quad P(D) = 0.08 + 0.02 = 0.10.$$

$$C \cap D = \{FF\}, \quad P(C \cap D) = 0.02.$$

Adding $P(C) + P(D)$ includes the FF probability twice.

Deriving the addition rule

In the component-inspection model, C is first-component failure and D second-component failure. Use $P(C)=0.20$, $P(D)=0.10$ and $P(C \cap D)=0.02$ to find $P(C \cup D)$.

Subtract the overlap once:

$$P(C \cup D) = P(C) + P(D) - P(C \cap D).$$

For the inspection model,

$$P(C \cup D) = 0.20 + 0.10 - 0.02 = 0.28.$$

This agrees with the direct calculation of at least one failure.

Practice: overlapping system alerts

Assume $P(A) = 0.20$, $P(B) = 0.15$ and $P(A \cap B) = 0.08$ for two system alerts. Find the probability of any alert.

$$P(A \cup B) = 0.20 + 0.15 - 0.08 = 0.27.$$

The overlap is already included in each marginal probability.

The probability of no alert is

$$1 - 0.27 = 0.73.$$

Practice: exactly one system alert

Assume $P(A)=0.20$, $P(B)=0.15$ and $P(A \cap B)=0.08$ for two system alerts. Calculate the probability of exactly one alert.

Only A :

$$P(A \setminus B) = 0.20 - 0.08 = 0.12.$$

Only B :

$$P(B \setminus A) = 0.15 - 0.08 = 0.07.$$

These cases are disjoint, so the required probability is $0.12 + 0.07 = 0.19$.

Equally likely outcomes

A batch contains 20 items, of which 3 are defective. Select one item uniformly. Find the defect probability.

There are 20 distinct item identifiers, each with probability $1/20$.

The defective-item event contains three identifiers:

$$P(\text{defective}) = 3 \times \frac{1}{20} = \frac{3}{20}.$$

Counting favourable outcomes works here because uniform selection makes the item identifiers equally likely.

A die with unequal probabilities

Suppose a die has $P(\{k\}) = k/21$ for $k = 1, \dots, 6$. Find the probability of an even face.

The weights are valid because $1 + 2 + 3 + 4 + 5 + 6 = 21$.

The even-face event is $\{2, 4, 6\}$, so

$$P(\text{even}) = \frac{2}{21} + \frac{4}{21} + \frac{6}{21} = \frac{4}{7}.$$

Three of the six faces are even, but their total probability is not $1/2$.

Practice: checking a proposed model

Three mutually exclusive outcomes exhaust an experiment. Proposed probabilities are 0.2, 0.3 and 0.6. Is the model valid?

The sum is $0.2 + 0.3 + 0.6 = 1.1$, so the model is invalid.

If the first two probabilities are fixed, the third must be

$$1 - 0.2 - 0.3 = 0.5.$$

Normalization is a requirement, not an approximation.

The sample space in a complete model

For one inspection, record 0 for pass and 1 for failure. Specify the possible outcomes.

$$\Omega = \{0, 1\}.$$

This set lists the possible outcomes. Outcome 1 means failure; outcome 0 means pass.

The sample space does not by itself specify how likely the two outcomes are.

The collection of events

For the inspection recorded as 0 = pass and 1 = failure, list every subset of $\Omega = \{0, 1\}$.

$$\mathcal{F} = \{\emptyset, \{0\}, \{1\}, \Omega\} = 2^\Omega.$$

These are the impossible event, pass, failure, and the certain event.

Here $1 \in \Omega$, while $\{1\} \in \mathcal{F}$: an outcome and an event are different objects.

The probability assignment

Suppose a pass/fail inspection has failure probability 0.10. Using 0 = pass and 1 = failure, assign a probability to every event.

Event	Probability
$\emptyset$	0
$\{0\}$	0.90
$\{1\}$	0.10
Ω	1

The probability measure P assigns probabilities to events.

The probability triplet

The example has outcomes $0 = \text{pass}$ and $1 = \text{failure}$, all four events, and failure probability 0.10 . Write the probability-space triplet.

$$(\Omega, \mathcal{F}, P)$$

$\Omega = \{0, 1\}$ lists the possible outcomes.

$\mathcal{F} = 2^\Omega$ lists the events.

P assigns a probability to each event.

These three objects form a probability space.

Why the event collection must be closed

If we can ask whether A occurs, what related questions should the model also describe?

Whether A does not occur: the complement A^c .

Whether at least one of $A_1, A_2, \dots$ occurs: the union $\bigcup_n A_n$.

An event collection containing Ω and closed under complements and countable unions is a **σ -algebra**.

For a finite model we ordinarily use all subsets; this collection satisfies these requirements.

The probability rules

Complements let us ask what cannot occur; countable unions let us ask whether at least one of a list of events occurs. Once these questions are events, what rules should their probabilities obey?

Kolmogorov's axioms constrain P on the event collection $\mathcal{F}$:

- 1. **Nonnegativity:** $P(A) \geq 0$ for every $A \in \mathcal{F}$.
- 2. **Normalization:** $P(\Omega) = 1$; an experiment produces an outcome in Ω .
- 3. **Countable additivity:** if $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint, then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

The union is “at least one”; disjointness prevents double counting. Complements and countable unions define the event collection; these three axioms govern its probabilities.

Running frequency in a model

In the independent fair-coin model, the lab plots the running fraction of heads beside the model probability p . What can a finite run illustrate? What does it establish about a limit?

In this stipulated model, each toss is independent, and a head has probability $p = 1/2$.

The plot shows the realized proportion after each finite number of tosses beside the model probability.

A finite simulation illustrates sample variation. By itself, it does not prove a limiting theorem.

Practice: a batch inspection triplet

Ten labelled items contain two defective items, numbered 3 and 8. Select one item uniformly. Specify the triplet and the defect event.

$$\Omega = \{1, 2, \dots, 10\}, \quad \mathcal{F} = 2^\Omega.$$

For each event A , $P(A) = |A|/10$.

The defect event is $D = \{3, 8\}$, hence $P(D) = 2/10$.

A different selection rule would require a different probability assignment.

Recording a numerical result

For the inspection model, the probabilities of PP, PF, FP and FF are 0.72, 0.08, 0.18 and 0.02. Let X count failed components. Assign X to each outcome.

Outcome ω	PP	PF	FP	FF
$X(\omega)$	0	1	1	2

X is a function: it assigns one number to each outcome. Several outcomes may have the same numerical value.

The event $X = 1$

For the ordered inspection, X counts failures and $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$. Find $P(X=1)$.

$$\{X = 1\} = \{PF, FP\}.$$

Under the stated model,

$$P(X = 1) = 0.08 + 0.18 = 0.26.$$

First identify the outcomes; then add their probabilities.

The distribution of the failure count

For the ordered inspection, X counts failures and $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$. Find its probability distribution.

x	Outcomes producing x	$P(X = x)$
0	PP	0.72
1	PF, FP	0.26
2	FF	0.02

Check: $0.72 + 0.26 + 0.02 = 1$.

The three count values are not equally likely.

A numerical set and its inverse image

For PP, PF, FP, FF with probabilities 0.72, 0.08, 0.18, 0.02, let X count failures. For $B=\{1,2\}$, find $X^{-1}(B)$ and $P(X \in B)$.

$$X^{-1}(B) = \{\omega: X(\omega) \in B\} = \{PF, FP, FF\}.$$

This is the inverse image of B .

$$P(X \in B) = 0.08 + 0.18 + 0.02 = 0.28.$$

An inverse image converts a numerical condition to an event in the original experiment.

An inverse image need not be an inverse function

On $\Omega = \{PP, PF, FP, FF\}$, X counts failures. Find $X^{-1}(\{1\})$ and explain whether an outcome can be recovered uniquely from $X=1$.

No: either PF or FP could have occurred.

X has no inverse function on these outcomes.

Nevertheless, the inverse image is well defined:

$$X^{-1}(\{1\}) = \{PF, FP\}.$$

A random variable and a probability measure

For outcomes PP, PF, FP, FF with probabilities 0.72, 0.08, 0.18, 0.02, let X count failures. Find $X(PF)$ and $P(\{PF, FP\})$. What does each map take as input?

Despite its name, X is not itself a random rule. Once the experiment and recording rule are fixed, X is a fixed function on the set of outcomes. The random outcome supplied to it gives its value a distribution. It maps each outcome to a real number:

$$X: \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto X(\omega).$$

Precisely, X is measurable when, for every Borel set $B \subseteq \mathbb{R}$, the inverse image

$$X^{-1}(B) = \{\omega \in \Omega: X(\omega) \in B\}$$

belongs to $\mathcal{F}$. Here $X(PF) = 1$.

Probability is the set function $P: \mathcal{F} \rightarrow [0, 1]$: it maps an event to a number in the unit interval. Thus

$P(\{PF, FP\}) = 0.08 + 0.18 = 0.26$. More generally, $P_X(B) = P(X^{-1}(B))$. In this finite model, $\mathcal{F} = 2^\Omega$, so every real-valued function on Ω is measurable.

Practice: a repair-cost random variable

For the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$, let X count failures and $Y=500X$ rupees. Find $P(Y>500)$.

The possible costs are ₹0, ₹500 and ₹1000.

$$\{Y > 500\} = \{X = 2\} = \{FF\}.$$

Therefore $P(Y > 500) = 0.02$.

For comparison, $P(Y \geq 500) = P(X \geq 1) = 0.28$. The strict inequality matters.

A uniform measurement model

Let E be a uniform measurement error on $[-2, 2]$ mm. State how to calculate the probability of an interval $[a, b]$ contained in this range.

Intervals of the same length have the same probability.

For an interval contained in $[-2, 2]$,

$$P(a \leq E \leq b) = \frac{b - a}{4}.$$

The model assigns probability by interval length.

Meeting a tolerance

Assume measurement error E is uniform on $[-2, 2]$ mm. Find the probability that the absolute error is at most 0.5 mm.

$$|E| \leq 0.5 \Leftrightarrow -0.5 \leq E \leq 0.5.$$

The acceptable interval has length 1 mm; the full interval has length 4 mm.

$$P(|E| \leq 0.5) = \frac{1}{4}.$$

For this continuous model, including or excluding either endpoint does not change the probability.

A uniform source variable

Let U be uniform on $(0,1)$. Find $P(U < 0.25)$ and $P(U > 2/3)$.

$$P(U < 0.25) = 0.25.$$

$$P(U > 2/3) = 1 - 2/3 = 1/3.$$

We will generate measurements from U and translate their events back to intervals of U .

Transforming a uniform variable

Let U be uniform on $(0,1)$ and $D=10U$ cm. Find $P(D<3 \text{ cm})$.

$$D < 3 \Leftrightarrow 10U < 3 \Leftrightarrow U < 0.3.$$

Thus $P(D < 3) = 0.3$.

The inverse transformation is $U = D/10$.

This change of scale gives a uniform distance on $(0, 10)$ cm.

A nonlinear transformation

Let U be uniform on $(0,1)$ and $D=10\sqrt{U}$ cm. Find $P(D<3$ cm).

$$D < 3 \Leftrightarrow \sqrt{U} < 0.3 \Leftrightarrow U < 0.09.$$

Thus $P(D < 3) = 0.09$.

The inverse transformation is $U = (D/10)^2$.

The possible distances are the same as before, but their probabilities differ.

Sampling records from two sites

A hypothetical dataset has 80 records from site A and 20 from site B. Select one of the 100 records uniformly. Find the probability that it came from each site.

Each of the 100 records has probability $1/100$.

$$P(\text{site A}) = 80/100 = 0.8.$$

$$P(\text{site B}) = 20/100 = 0.2.$$

The site labels themselves are not equally likely.

Selecting a site before selecting a record

A hypothetical dataset has 80 records from site A and 20 from site B. Select A or B with probability $1/2$ each, then uniformly within the selected site. Find the selection probability of one specified record from each site.

A particular record from A:

$$\frac{1}{2} \times \frac{1}{80} = \frac{1}{160}.$$

A particular record from B:

$$\frac{1}{2} \times \frac{1}{20} = \frac{1}{40}.$$

Each B record is four times as likely as each A record. The factors describe the two stages of the stated sampling procedure.

Accuracy and class proportions

A hypothetical test set contains 9,900 nondefective and 100 defective items. A system always predicts “nondefective.” Calculate its accuracy and defect detection fraction.

Accuracy:

$$\frac{9900}{10000} = 0.99.$$

Fraction of defective items detected:

$$\frac{0}{100} = 0.$$

The overall proportion correct does not specify performance within the defective class.

Practice: two fair dice

Assume two independent fair dice. List the 36 equally likely ordered outcomes and find the probability that the sum is 7.

There are 36 ordered pairs, each with probability $1/36$ under the stated assumptions.

The favourable pairs are

$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$.

Thus $P(\text{sum } 7) = 6/36 = 1/6$.

Practice: at least one six

Assume two independent fair dice. Using the complement, find the probability of at least one six.

The complement is “neither die shows six.”

There are $5 \times 5 = 25$ such ordered pairs.

$$P(\text{at least one six}) = 1 - \frac{25}{36} = \frac{11}{36}.$$

The favourable outcomes are not just (6, 6).

Review: describing a probability calculation

A hypothetical log has 50 requests, 7 above 200 ms. Select one request uniformly. Define the outcome, the latency random variable, and the event of latency above 200 ms; calculate its probability.

Outcome: a request identifier in $\Omega = \{1, \dots, 50\}$.

Random variable: $X(i)$ is the recorded latency of request i , in milliseconds.

Event: $A = \{i: X(i) > 200\}$, containing seven identifiers.

With all subsets as events and uniform selection,

$$P(A) = 7/50 = 0.14.$$

Review: assumptions behind the answer

For a probability calculation based on data or a transformed random variable, what must be specified about the experiment, event and distribution?

Specify the experiment and the sampling rule.

Write the event as a set or an equivalent numerical condition.

Use probabilities consistent with the sampling rule.
Counting requires equally likely outcomes.

For a transformation, carry the distribution with it and check the inverse image of the event.

Further reading

Use the main notes for additional worked problems and the supplement for the formal development.

Jay L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th edition, Chapters 2–4: sample spaces and events, probability, random variables and their distributions. The **publisher's table of contents** is provided for reference.

Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, Chapters 1, 4, and 11. The examples here are stipulated for this course.



Speaker notes